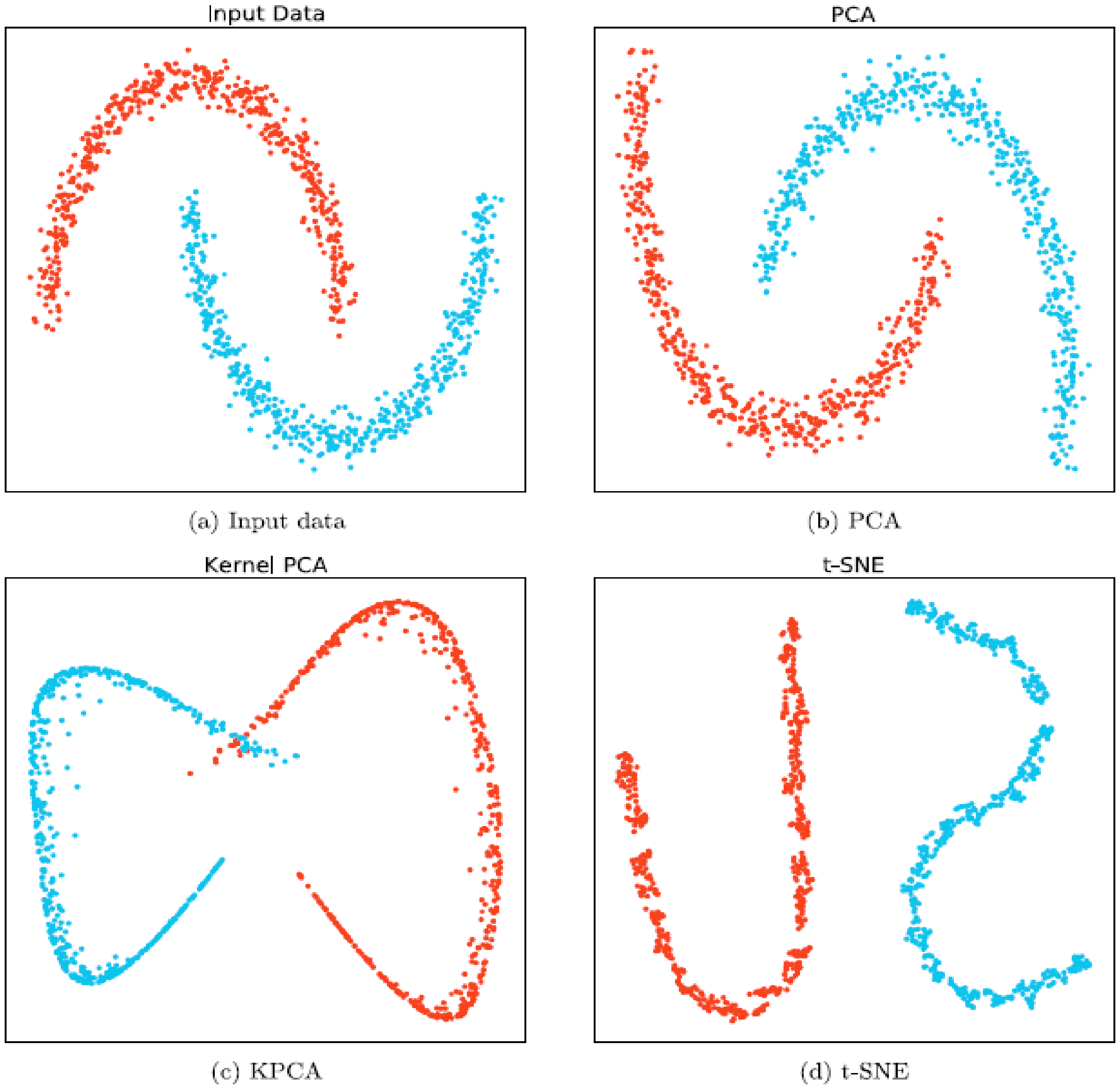
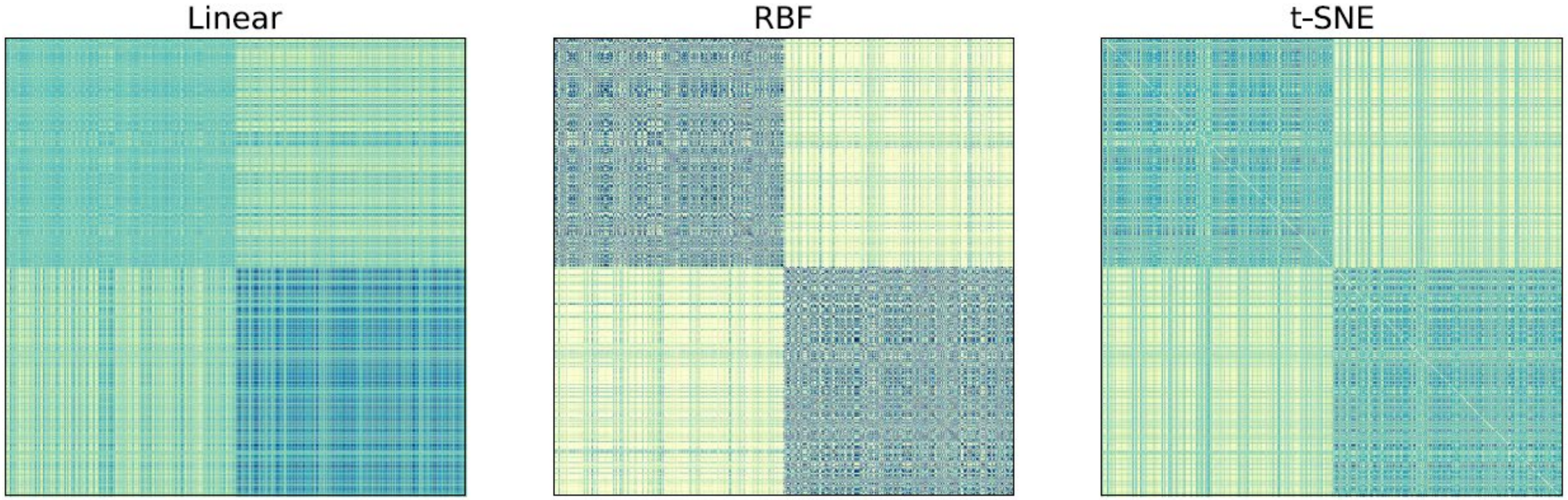
Example: Comparison on Nonlinear Data Points: Linear vs. Nonlinear Dimensional Reduction Methods

- Visualization: An example of linear vs. nonlinear dimensionality reduction methods
  - Given a collection of input data in 2-D space (Fig. (a)): Red and blue data points are not linearly separable
  - PCA transformation can not make it linearly separable
  - KPCA can make the points linearly separable
  - t-SNE (t-distributional NSE) can make them linearly separable



(a) Input data (b) PCA (c) KPCA (d) t-SNE

Heatmap of the Proximity Matrices: Linear vs. Nonlinear Dimensional Reduction Methods



(a) PCA (b) KPCA t-SNE

The heatmaps of the proximity matrices in PCA (a), KPCA (b), and t-SNE(c)

- The two diagonal blocks indicate the proximity within the two clusters respectively
- The two off-diagonal blocks indicate the proximity between the data from the two clusters
- With nonlinear methods (KPCA and t-SNE), the proximity between data tuples from the same cluster is much higher than the proximity between data tuples from different clusters